13.3 Architecting for high availability

Before you begin implementing highly available or even fault-tolerant architectures on AWS, you should start by analyzing your disaster-recovery requirements. Disaster recovery is easier and cheaper in the cloud than in a traditional data center, but building for high availability increases the complexity and, therefore, the initial costs as well as the operating costs of your system. The recovery time objective (RTO) and recovery point objective (RPO) are standards for defining the importance of disaster recovery from a business point of view.

Recovery time objective (RTO) is the time it takes for a system to recover from a failure; it's the length of time until the system reaches a working state again, defined as the system service level, after an outage. In the example with a Jenkins server, the RTO would be the time until a new virtual machine is started and Jenkins is installed and running after a virtual machine or an entire availability zone goes down.

Recovery point objective (RPO) is the acceptable data-loss time caused by a failure. The amount of data loss is measured in time. If an outage happens at 10:00 a.m. and the system recovers with a data snapshot from 09:00 a.m., the time span of the data loss is one hour. In the example of a Jenkins server using autoscaling, the RPO would be zero, because data is stored on EFS and is not lost during an AZ outage. Figure 13.8 illustrates the definitions of RTO and RPO.

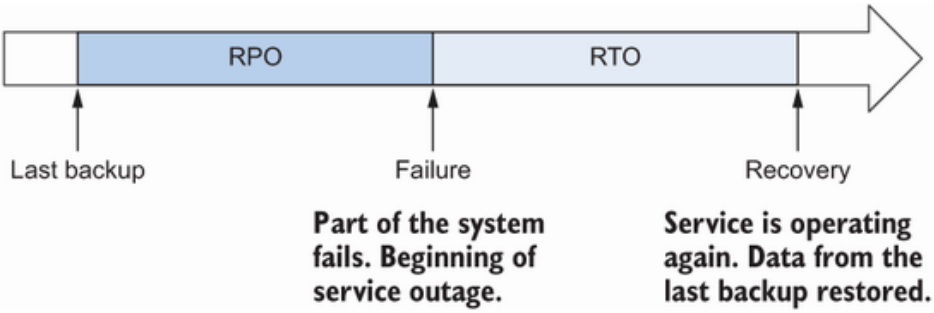


Figure 13.8 Definitions of RTO and RPO